Raj Jung Mahat

Ph.D. Candidate, Mechanical Engineering

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Summary

Materials-informatics researcher and data scientist with five years of experience turning noisy, high-throughput mechanical-test data into validated, uncertainty-aware material models. Ph.D. candidate at Georgia Tech (expected 2027) in the HTMDEC / BIRDSHOT high-throughput materials-discovery program on FCC high-entropy alloys (HEAs): Bayesian inference of elastic–plastic constitutive laws from small punch tests (SPT), finite-element simulation, Gaussian-process surrogates, identifiability analysis and model-discrepancy modeling, validated across nine alloys and HTMDEC HEA campaigns. At Multiscale AI, built production ML workflow modules, an inverse-inference toolkit (surrogates + MCMC) and a Cox survival model for adhesive-bond durability (Novelis).

Core skills

Materials informatics	High-throughput mechanical testing (small punch, tensile, DIC), constitutive modeling (Voce isotropic and Armstrong–Frederick kinematic hardening), composition–property and process–structure–property data, EBSD-informed analysis of HEA batches
Statistics & ML	Bayesian inference, Gaussian processes, MCMC (transitional, Hamiltonian, Riemannian), normalizing flows, survival analysis (Cox PH), PCA, diffusion maps, sensitivity and identifiability analysis, design of experiments, active learning, uncertainty quantification
Simulation	Abaqus finite-element modeling incl. UMAT development; FEM, numerical methods for PDEs
MLOps & tools	Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Pyro, PyMC, lifelines), SQL, R, MATLAB, Fortran, LaTeX; inference/data/feature-engineering pipelines, logging, Git, Linux and HPC batch workflows

Experience

2022–present **Graduate Research Assistant**, *Georgia Institute of Technology*, Atlanta, GA

Advisor: Prof. Surya Kalidindi. Materials informatics and Bayesian calibration from high-throughput mechanical tests within the HTMDEC / BIRDSHOT program on FCC high-entropy alloys.

- **HTMDEC high-entropy alloys.** First author of a Bayesian framework (simulation surrogate + model-discrepancy correction) that infers Voce elastic–plastic parameters from SPT load–displacement curves of HTMDEC/BIRDSHOT Al–V–Cr–Mn–Fe–Co–Ni–Cu HEAs, benchmarked against tensile tests.
- **Ongoing: kinematic hardening of HEAs in SPT.** Characterizing back-stress (kinematic-hardening) response of HTMDEC HEAs from small punch tests, beyond isotropic Voce hardening.
- **HEA data analysis.** SPT, tensile, EBSD and composition data across eight vacuum-arc-melted HEA batches; simulated ∼60 HEAs and linked the systematic SPT simulation–experiment mismatch to grain size.
- **Constitutive calibration and discrepancy.** Across nine alloys (Al 2024/5052/6061/7075, Cu, brass, Inconel 718, nickel silver, SS316), one fixed discrepancy direction captures 93.7% of the SPT–tensile mismatch; a response-dependent discrepancy model in latent PC space cut cross-test error from 28% to 13%.
- **GP surrogates and MCMC.** Gaussian-process emulators (ARD kernels) on a PCA-compressed response space replace finite-element solves inside the likelihood; transitional and Hamiltonian MCMC for multimodal posteriors.
- **Identifiability.** Likelihood-informed-subspace analysis separating parameter directions a test resolves from those it leaves unconstrained; used to guide test design.
- **Experiments.** Designed a camera-instrumented (DIC) small punch rig for full-field strain; contributed experiments to an SPT + DIC amortized-inference study.
- **Mentoring and teaching.** Mentored undergraduates on a DIC project; TA for *Machine Learning for Mechanical Engineers* (Spring 2024).

Jan–May 2025 **Data Scientist**, *Multiscale AI*, Atlanta, GA

Production ML workflow development and applied survival modeling for industrial materials clients.

- Designed and implemented MLOps modules for the company's ML platform (inference workflow, data workflow, feature-engineering pipeline) with a logging backend for run traceability.
- Designed an inverse-inference workflow module (surrogate-model design, MCMC sampling) packaged as a reusable toolkit across client problems.
- Built a Cox proportional-hazards model of adhesive-bond durability for a Novelis aluminum surface-treatment study: time-to-failure under humidity vs. laser-treatment parameters and surface measurements (roughness, oxide ratio), with right-censored observations.

2018–2021 **Graduate Research Student**, *Indian Institute of Science (IISc)*, Bangalore, India

Data-driven extraction of uniaxial creep properties from instrumented indentation: trained neural-network inverse models that map indentation response to material parameters.

Selected projects

Sampling & optimization — Leapfrog HMC from scratch with a convergence study (acceptance and mixing vs. step size and trajectory length) on ODE-defined posteriors; Riemannian-manifold MCMC benchmarked against HMC on strongly correlated posteriors; normalizing flows compared with MCMC on quality and cost.

Latent-space modeling — PCA, graph-Laplacian and diffusion-map embeddings of high-dimensional test responses; response-dependent corrections conditioned on the latent coordinates of the response itself.

Gaussian measures — Open learning notes and notebooks on the Feldman–Hájek theorem and SPDE priors (equivalent vs. singular Gaussian measures under mesh refinement): rajjung421.github.io/gaussian-measure-equivalence.

Education

2022–2027 (exp.) **Ph.D., Mechanical Engineering**, *Georgia Institute of Technology*, Atlanta, GA

Doctoral minor: Mathematics. Dissertation: Bayesian constitutive calibration and identifiability under compressed mechanical observation. Coursework: Numerical Linear Algebra, Numerical Methods for PDEs and ODEs, Multivariate Statistics, Statistical Methods, Advanced Design of Experiments, Computational Data Analysis, Finite Element Method, Micromechanics.

2018–2021 **M.S., Materials Engineering**, *Indian Institute of Science*, Bangalore, India

2013–2017 **B.E., Mechanical Engineering**, *M.S. Ramaiah Institute of Technology*, Bangalore, India

Publications and manuscripts

2026 **R. J. Mahat**. Measure equivalence as a guide for learning SPDE priors from replicated fields. Preprint. doi:10.5281/zenodo.23443597

in prep. **R. J. Mahat**, A. Venkatraman, S. R. Kalidindi. Bayesian calibration of Voce hardening parameters from the small punch test: identifiability and structured cross-test discrepancy.

under review S. Mahmoud, A. Venkatraman, **R. J. Mahat**, et al., S. R. Kalidindi. Amortized posteriors for estimation of material constitutive parameters from multimodal measurements on small punch tests.

2026 N. Flores, et al. (incl. **R. Mahat**, S. R. Kalidindi, R. Arróyave). Data-driven insights into composition–property relationships in FCC high entropy alloys. *Scripta Materialia* 274, 117136. doi:10.1016/j.scriptamat.2025.117136

2024 M. Mulukutla, A. N. Person, S. Voigt, et al. Illustrating an effective workflow for accelerated materials discovery. *Integrating Materials and Manufacturing Innovation* 13, 453–473. doi:10.1007/s40192-024-00357-3